Pre-specified statistical analysis plan

Variability of life-history parameters of penaeid shrimps exploited in Brazil: among species, regions, research groups and estimation methods

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This document complements the methods document (search strategy, screening and extraction) and is deposited together with it.

1. Statement on the timing of registration

The literature search and the data extraction were completed **before** this registration (extraction began in August 2026). The eligibility criteria were not changed after extraction began. Before registration the data were examined only descriptively (counts, cross-tabulations, ranges and quality control); no inferential model was fitted with the aim of testing the hypotheses below. The hypotheses, models and decision rules of this version were fixed on the basis of the structure of the database (number of estimates, studies, populations and groups per stratum), which is known, and not on the results.

This is a **post-collection, pre-analysis** registration. OSF time-stamps it, and any later change is made through a new registered and justified version (Section 10), so that the reader can distinguish what was planned from what was explored.

2. Objective and unit of analysis

To quantify the variability of life-history parameters (growth, size at maturity) of five penaeid shrimp species of Brazil and to **partition that variability** among species, latitude, research groups, estimation methods, studies and populations. The unit of analysis is the **estimate** (one row of the master spreadsheet: a value as published, for one sex, method and locality). The database has 1,322 estimates from 162 sources, 150 primary studies and 184 populations, each identified by estimate_id, primary_study_id, population_id and research_group. A population is the set of individuals of one species from the same locality and sampling period; rows sharing a population_id come from the same field sample (article, thesis, report or re-analysis) and are not independent. Rows are labeled main, sensitivity, redundant or exclude (column analysis_use) by mechanical rules described in the methods document: redundant rows repeat all parameters of another publication of the same study, sensitivity rows are secondary citations (reliability levels 4–5) or use a length measure that is not comparable, and exclude rows contain no sampled population.

3. Hypotheses

The primary response is the von Bertalanffy growth coefficient k, standardized to year^{-1} (month × 12, day × 365). Because k does not depend on the length measure, it is analyzed with all length measures together. Growth performance ($\varphi' = \log10\ k + 2\ \log10\ L\infty$) and size at 50% maturity (L50) depend on the length measure and are analyzed separately for carapace length (CL) and total length (TL), without conversion between them. $L\infty$ is not analyzed on its own; it is described by species and length measure. Any analysis not listed in Table 1 is declared exploratory.

Table 1. Confirmatory hypotheses, responses and effects structure.

Hypothesis	Response	Fixed effects	Random effects
H1. Species differ in growth. Expected direction: species differ in growth, largely because of the estimation method (method class explains a large share of the variation among estimates).	log k (primary); φ' (secondary, by length measure)	species; method class; sex; latitude	research group, study, population

Hypothesis	Response	Fixed effects	Random effects
H2. Growth does not vary with latitude once research group and method are accounted for. Expected result: the latitude slope is practically null, i.e. its 95% credible interval lies within ±10% change in k per 10° of latitude (Section 5.3).	log k; φ′	latitude (per 10°, positive southward); species; method class; sex	research group, study, population
H3. The estimation method changes the estimated k (modal progression, ELEFAN/frequency analysis, linearization/regression, direct age). No direction is specified.	log k	method class; species; sex; latitude	research group, study, population
H4. Variance in growth is attributable to research group, study and population beyond the residual variance (variance partitioning, not a test).	log k; φ′	none (null model), then the full model of H1	research group, study, population
H5. L50 differs between sexes (females larger).	log L50 (CL; TL)	sex; species; latitude	research group, study, population

Exploratory analyses declared in advance: (E1) association of log k with the year of data collection, controlling for the same fixed and random effects as H1; (E2) the latitude slope estimated for Xiphopenaeus kroyeri alone. Natural mortality M and total mortality Z are not used as responses (Section 4).

4. Data available per analysis

Table 2 shows how many estimates, studies, populations and research groups each analysis has, considering only rows with analysis_use = main and excluding the values flagged in repeated_parameters. It was obtained from the master spreadsheet and precedes the running of any model.

Table 2. Sample size per analysis (main rows).

Analysis	Estimates	Studies	Populations	Groups	Note
k, all length measures	306	38	64	16	X. kroyeri: 125 estimates, 34 populations
k with a defined method class (model data)	296	37	59	16	excludes 10 rows of "not reported/other"
φ′ in CL	182	29	49	11	F. subtilis: 4 estimates
φ′ in TL	131	15	21	12	L. schmitti: 4 estimates
L50 in CL	96	39	55	14	X. kroyeri: 65; F and M in the same population: 16 populations
L50 in TL	50	28	26	14	F and M in the same population: 11 populations

Four consequences of this table are decisions of the plan, made before seeing results. **First**: k is the primary response because it has the largest sample and does not depend on the length measure; φ′ is analyzed within each length measure and requires L∞ and k from the same row. **Second**: L∞ is not a response, because it is reported in CL (244 rows) or TL (193 rows) and only 16 rows carry both. **Third**: M and Z are not analyzed. Of the 165 main rows with M, almost all values come from empirical equations (Pauly 1980, Taylor 1960, Rikhter and Efanov 1976 and others) that are functions of k, L∞, longevity or temperature, so treating them as independent information would create pseudoreplication; Z depends on fishing mortality and is not a property of the population alone. **Fourth**: species with few populations (F. brasiliensis 5, F. subtilis 6, F. paulensis 7, L. schmitti 12 contributing k, against 34 for X. kroyeri) are not analyzed in separate models; all species enter one model.

5. Models

5.1. Model structure

Each response is analyzed with a Bayesian multilevel model with Gaussian errors on the log scale (φ′ is already logarithmic). All models have the same structure so that effects are comparable among hypotheses: species, sex and latitude as fixed effects, plus estimation-method class for the growth responses, and random intercepts for research group, primary study and population. Study is nested in research group and population in study, with two exceptions (one study with estimates attributed to two groups and two populations that appear in two studies); random intercepts are therefore specified with unique identifiers at each level rather than with nested syntax. Sex enters with levels F, M and pooled. Latitude (lat10) is the decimal latitude of the population divided by 10, with positive values southward, centered on its mean. The reference level of method class is modal progression, the most frequent class.

```
brm(log_k ~ species + sex + method + lat10 +
        (1|research_group) + (1|primary_study_id) + (1|population_id),
    family = gaussian(),
    prior = c(prior(normal(0, 1), class = b), prior(student_t(3, 0, 1), class = sd),
              prior(student_t(3, 0, 1), class = sigma)),
    chains = 4, iter = 4000, control = list(adapt_delta = 0.95))
```

For φ′ and L50 the response is replaced and the model is fitted separately in the CL and TL strata (the method term is omitted for L50). Convergence criteria: R-hat ≤ 1.01, effective sample size ≥ 400 and no divergent transitions; posterior predictive checks are shown for each model. If divergent transitions occur, adapt_delta is raised to 0.99 and this is reported; the random-effects structure is not simplified.

5.2. Variance partitioning and contribution of each factor (H4)

A null model with only the three random intercepts gives the proportion of total variance at each level (research group, study, population and residual; variance-partition coefficients). The full model is then fitted, and two quantities are reported with credible intervals: (i) the Bayesian R^2 of the fixed effects (marginal) and of fixed plus random effects (conditional); and (ii) the change in each variance component, and in the expected log predictive density (loo), when a fixed term (species, method class, latitude) is removed. The contribution of the estimation method (H3) is therefore reported as an effect size (difference in log k between method classes, with its interval), as the variance explained by the method term, and as the change in loo, and not as a test based on an information criterion alone. Because research group, region and method overlap in the database, the unique contribution of each fixed term is reported together with the contribution that it shares with the others (Section 7).

5.3. Inference and the equivalence margin for H2

Fixed effects are reported as posterior medians with 95% highest-density credible intervals and the probability of direction. Contrasts among species include all pairs (H1), and the contrasts among method classes include all pairs (H3). Variance components are reported as standard deviations and as proportions of the total variance.

For H2, the absence of an effect is assessed with a **region of practical equivalence** defined before the analysis: a change smaller than ±10% in k per 10° of latitude is considered biologically irrelevant. On the natural-log scale the margin is ±0.095 (ln 1.1); for φ′ (log10 scale) it is ±0.041 (log10 1.1, a 10% change in k at constant L∞). The latitude slope is classified as **practically null** if its whole 95% credible interval lies inside the margin, **inconclusive** if the interval crosses the margin, and **different from zero** if the interval lies wholly outside it. The share of the posterior distribution inside the margin is also reported. An interval that includes zero but crosses the margin is not interpreted as absence of an effect. The margin was fixed on biological grounds and before any model was fitted.

6. Exclusions and data rules

- Only rows with analysis_use = main enter the main analyses; redundant, sensitivity and exclude rows are left out (Section 8).
- Parameters flagged in repeated_parameters are omitted from the per-parameter analyses; the cited row keeps the value. When two works report different values for the same population (for example a re-analysis of the same sample), both are kept and the population random intercept absorbs their non-independence.
- k is converted to year^{-1} (month × 12, day × 365). Values of L∞ fixed by convention, M calculated by empirical equations and longevity calculated as 4.605/k are not used as independent responses.
- φ′ in CL uses only rows with cl_definition = standard (assumed); rows with a rostrum-based definition awaiting verification are excluded. Ferreira (2013), with a post-orbital measure, is kept for the sensitivity analysis.

- The 10 growth rows with method class "not reported/other" are excluded from models that include method class.
- Missing data: complete-case analysis for each model; there is no imputation.

7. Identifiability diagnostics

Research group, study and method overlap in the database. For k, 16 groups contribute; 9 of them contribute a single study, one group (NEBECC-UNESP) contributes 11 of the 38 studies, and 12 of the 16 groups operate in a single macro-region. In these groups, group and study cannot be separated, and the group variance is estimated mainly from the groups with several studies. Only 8 of the 38 studies applied more than one growth method and 11 populations have estimates from more than one method, so most of the method effect is a comparison among studies. Method class is also related to time: only 8 of the 131 modal-progression estimates with a known year date from 2000 or earlier. Before each model is interpreted, the following will be reported: cross-tabulations of group × region × species × method, the rank of the design matrix, and the sensitivity of the estimates to alternative priors (sd ~ exponential(2) and half-normal(0, 0.5)). If a fixed effect is not identifiable with the random effects of the model, it will be declared **not estimable**, and the group random effect will not be removed to make it estimable. The wide credible intervals expected from 64 populations and 16 groups are reported as such.

8. Sensitivity analyses

- Inclusion of the rows flagged as sensitivity (secondary citations of reliability levels 4 and 5 and the post-orbital length measure), and exclusion of rows with a quality-control flag.
- One estimate per population (the one with the largest n) versus all.
- Journal articles versus grey literature (theses, dissertations, reports).
- Removal, one at a time, of each research group (leave-one-group-out), and removal of the 9 groups that contribute a single study.
- Alternative priors for the standard deviations of the random effects (Section 7).

Limitation declared in advance: records of Xiphopenaeus kroyeri may include more than one species, because the source works do not distinguish them (taxonomic uncertainty before 2019). This heterogeneity is already present in the published estimates, cannot be resolved by the statistical analysis of the parameters, and may inflate the residual variance of that species.

9. Validation of the extraction, use of AI and availability

Screening and extraction were carried out by two human researchers (M.S.S. and A.C.A.F.), independently: each located and read the works and transcribed the values; each validated their own extraction with a large language model (Claude, Anthropic, in separate accounts); and the two sets of results were then compared independently, with the article re-read whenever the values did not match. The comparison covered all sources (162), not a sample; there were one or two divergent sources, in both cases because of internal inconsistency of the work itself (estimates that do not hold across text, tables and figures), handled by the rule of keeping the published value and flagging the inconsistency. Agreement is reported as the number of divergent sources, not as Cohen's κ. The declaration of the use of computational assistance will accompany the manuscript according to the journal policy.

All analyses are run in R. The models are fitted with brms (Stan backend); posterior summaries and convergence diagnostics use the posterior and bayesplot packages, model comparison uses loo, and contrasts among species and among method classes use emmeans. R and package versions are recorded with sessionInfo() and deposited with the code.

The database (with the change-log, removed-rows and identifier sheets), the organization scripts, the analysis plan and the model code will be deposited with a DOI and a CC-BY license.

10. Changes to this plan

Any deviation (model, exclusion, hypothesis) will be recorded as a new version of this document, with date and justification, and the manuscript will indicate which results are confirmatory, which are exploratory and which result from deviations.

References

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